

LAB 3: Hadoop – MeanMaxTemp

From the following link extract the weather data

<https://github.com/tomwhite/hadoopbook/tree/master/input/ncdc/all> a) Create a

MapReduce program to find average temperature for each year from NCDc data set. b) find the mean max temperature for every month.

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hadoop@bmscscce-HP-Pro-Tower-400-G9-PCI-Desktop-PC: $ start-all.sh
WARNING: Attempting to start all Apache Hadoop daemons as hadoop in 10 seconds.
WARNING: This is not a recommended production deployment configuration.
WARNING: Use CTRL-C to abort.
Starting namenodes on [localhost]
localhost: namenode is running as process 5420. Stop it first and ensure /tmp/hadoop-hadoop-namenode.pid file is empty before retry.
Starting datanodes
localhost: datanode is running as process 5562. Stop it first and ensure /tmp/hadoop-hadoop-datanode.pid file is empty before retry.
Starting secondary namenodes [bmscscce-HP-Pro-Tower-400-G9-PCI-Desktop-PC]
bmscscce-HP-Pro-Tower-400-G9-PCI-Desktop-PC: secondarynamenode is running as process 5809. Stop it first and ensure /tmp/hadoop-hadoop-secondarynamenode.pid file is empty before retry.
Starting resourcemanager
resourcemanager is running as process 6207. Stop it first and ensure /tmp/hadoop-hadoop-resourcemanager.pid file is empty before retry.
Starting nodemanagers
localhost: nodemanager is running as process 6381. Stop it first and ensure /tmp/hadoop-hadoop-nodemanager.pid file is empty before retry.
hadoop@bmscscce-HP-Pro-Tower-400-G9-PCI-Desktop-PC: $ jps
5809 SecondaryNameNode
14937 Jps
5562 DataNode
5420 NameNode
6381 NodeManager
6207 ResourceManager
hadoop@bmscscce-HP-Pro-Tower-400-G9-PCI-Desktop-PC: $ mkdir meannmax
hadoop@bmscscce-HP-Pro-Tower-400-G9-PCI-Desktop-PC: $ cd meannmax
hadoop@bmscscce-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~/meannmax$ nano MeanMaxDriver.java
hadoop@bmscscce-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~/meannmax$ nano MeanMaxMapper.java
hadoop@bmscscce-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~/meannmax$ nano MeanMaxReducer.java
hadoop@bmscscce-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~/meannmax$ mv *.java classes
hadoop@bmscscce-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~/meannmax$ javac -classpath $(hadoop classpath) -d classes *.java
Note: MeanMaxDriver.java uses or overrides a deprecated API.
Note: Recompile with -Xlint:deprecation for details.
hadoop@bmscscce-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~/meannmax$ jar -cvf meannmax.jar -C classes/ .
added manifest
adding: MeanMaxMapper.class(in = 1859) (out= 799)(deflated 57%)
adding: MeanMaxDriver.class(in = 1390) (out= 808)(deflated 41%)
adding: MeanMaxReducer.class(in = 1713) (out= 748)(deflated 56%)
hadoop@bmscscce-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~/meannmax$ wget https://raw.githubusercontent.com/tomwhite/hadoop-book/master/input/ncdc/all/1901.gz
--2026-05-04 14:30:57-- https://raw.githubusercontent.com/tomwhite/hadoop-book/master/input/ncdc/all/1901.gz
Resolving raw.githubusercontent.com (raw.githubusercontent.com)... 185.199.108.133, 185.199.110.133, 185.199.109.133, ...
Connecting to raw.githubusercontent.com (raw.githubusercontent.com)|185.199.108.133|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 73867 (72K) [application/octet-stream]
Saving to: '1901.gz'

1901.gz              100%[=====] 72.14K  --.-KB/s  in 0.03s

2026-05-04 14:30:57 (2.13 MB/s) - '1901.gz' saved [73867/73867]

hadoop@bmscscce-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~/meannmax$ wget https://raw.githubusercontent.com/tomwhite/hadoop-book/master/input/ncdc/all/1902.gz
--2026-05-04 14:30:57-- https://raw.githubusercontent.com/tomwhite/hadoop-book/master/input/ncdc/all/1902.gz
Resolving raw.githubusercontent.com (raw.githubusercontent.com)... 185.199.108.133, 185.199.110.133, 185.199.109.133, ...
Connecting to raw.githubusercontent.com (raw.githubusercontent.com)|185.199.108.133|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 73867 (72K) [application/octet-stream]
Saving to: '1902.gz'

1902.gz              100%[=====] 72.14K  --.-KB/s  in 0.03s

2026-05-04 14:30:57 (2.13 MB/s) - '1902.gz' saved [73867/73867]

hadoop@bmscscce-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~/meannmax$ gunzip -c 1901.gz 1902.gz > weather.txt
hadoop@bmscscce-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~/meannmax$ hdfs dfs -rm -r /meaninput
Deleted /meaninput
hadoop@bmscscce-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~/meannmax$ hdfs dfs -rm -r /meanoutput
Deleted /meanoutput
hadoop@bmscscce-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~/meannmax$ hdfs dfs -mkdir /meaninput
hadoop@bmscscce-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~/meannmax$ hdfs dfs -put weather.txt /meaninput
hadoop@bmscscce-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~/meannmax$ hadoop jar meanmax.jar MeanMaxDriver /meaninput /meanoutput
2026-05-04 14:32:39,823 INFO client.DefaultNoHARMAFailoverProxyProvider: Connecting to ResourceManager at /0.0.0.0:8032
2026-05-04 14:32:39,962 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the
2026-05-04 14:32:39,987 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/had
2026-05-04 14:32:40,151 INFO input.FileInputFormat: Total input files to process : 1
2026-05-04 14:32:40,244 INFO mapreduce.JobSubmitter: number of splits:1
hadoop@bmscscce-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~/meannmax$ hdfs dfs -cat /meanoutput/part-r-00000
01      4
02      1
03      6
04     34
05     90
06    144
07    184
08    172
09    123
10     73
11     21
12      3
hadoop@bmscscce-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~/meannmax$
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